

1. Determine the n th Taylor polynomial P_n for the function f .

(a) $f(x) = x - \cos x$.

(c) $f(x) = \sinh x$.

(b) $f(x) = e^{rx}$, r a real number.

(d) $f(x) = \ln(1 - x)$.

2. Let the function be

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0; \\ 0, & x = 0. \end{cases}$$

- Use L'Hôpital's rule to show that for every positive integer n

$$\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^n} = 0.$$

- Prove by induction that $f^{(n)}(0) = 0$ for all $n \geq 1$.
- What is the Taylor series of f ?
- For what values of x does the Taylor series of f actually converge to $f(x)$?

3. Find the Taylor polynomial of the function f for the given values of a and n and give the Lagrange form of the remainder.

(a) $f(x) = \arctan x$; $a = 1$ and $n = 5$.

(b) $f(x) = \ln x$; $a = 1$ and $n = 5$.

4. Expand $f(x)$ as indicated and specify the values of x for which the expansion is valid.

(a) $f(x) = (1 - 2x)^{-1}$ in powers of $x + 2$.

(b) $f(x) = \ln(1 + 2x)$ in powers of $x - 1$.

5. Find the interval of convergence

(a) $\sum \frac{(-1)^k}{\sqrt{k}} x^k$.

(d) $\sum \frac{7^k}{k!} x^k$.

(b) $\sum \frac{2^k}{k^2} x^k$.

(e) $\sum \left(\frac{k}{k-1} \right) \frac{(x+2)^k}{2^k}$.

(c) $\sum \frac{1}{(\ln k)^k} (x - 1)^k$.

(f) $\sum \left(1 + \frac{1}{k} \right) x^k$.

6. The power series $\sum_{k=0}^{\infty} a_k x^k$ has the property that $a_{k+3} = a_k$ for all $k \geq 0$.

(a) What is the radius of convergence of the series?

(b) What is the sum of the series?

7. Expand $f(x)$ in powers of x , basing your calculations on geometric series

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots$$

(a) $f(x) = \frac{1}{(1-x)^2}$.

(b) $f(x) = \ln(2 - 3x)$.

8. Sum the series

(a) $\sum_{k=0}^{\infty} \frac{1}{k!} x^{3k}.$

(b) $\sum_{k=0}^{\infty} \frac{1}{k!} x^{3k+1}.$

(c) $\sum_{k=0}^{\infty} \frac{3k}{k!} x^{3k-1}.$

9. Set $f(x) = xe^x$.(a) Expand $f(x)$ in a power series.

(b) Integrate the series and show that

$$\sum_{k=1}^{\infty} \frac{1}{n!(n+2)} = \frac{1}{2}.$$

10. Find the sum of the series

$$\sum_{k=1}^{\infty} a_k \quad \text{given that} \quad a_k = \int_k^{k+1} x e^{-x} dx.$$

11. Find the sum of the series

(a) $\sum_{k=0}^{\infty} \frac{(\ln 2)^k}{k!}.$

(b) $\sum_{k=0}^{\infty} \frac{3k}{k!} x^{3k-1}.$

12. Find a power series representation for the improper integral

(a) $\int_0^x \frac{\ln(1+t)}{t} dt.$

(b) $\int_0^x \frac{\arctan t}{t} dt.$

13. Find the Taylor series expansion of f and give the radius of convergence.

(a) $f(x) = \ln x$ in powers of $(x-1)$.

(c) $f(x) = \frac{2x}{1-x^2}$ in power of x .

(b) $f(x) = e^{-2x}$ in powers of $(x+1)$.

(d) $f(x) = \sin x^2$ in power of x .

14. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

(a) $\sum_{k=0}^{\infty} \frac{(-1)^k (100)^k}{k!}.$

(b) $\sum_{k=1}^{\infty} (-1)^{k-1} \frac{\ln k}{\sqrt{k}}.$

(c) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{(k+1)(k+2)}}.$